Team Number: 20

AI-POWERED SMART CLASSROOM:

SAFETY, ACCESS AND VISION CHALLENGE

Team Name: Do-Something

FastRobo 1.0

IEEE BUBT SB

BUBT

FastRoboX 1.0

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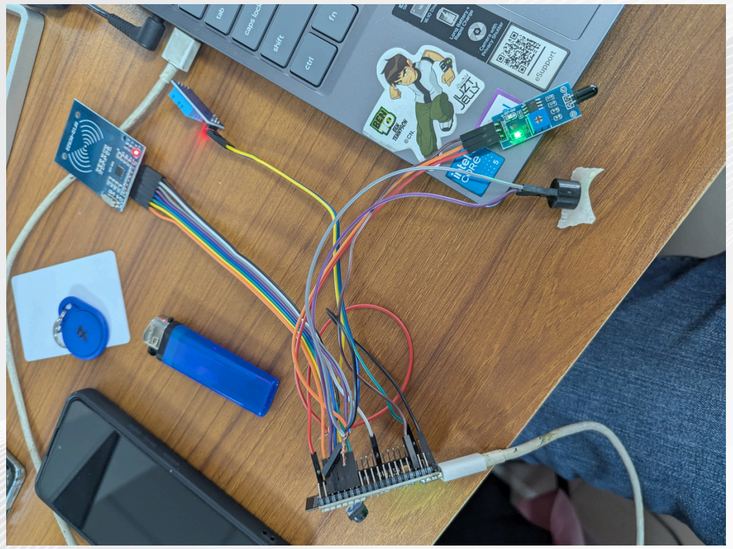
Introduction

FastRobox 1.0 is a 12-level challenge to build an AI-powered smart classroom using one ESP32-S3-CAM and the provided hardware kit. The system combines room safety monitoring, fire and panic response, secure access control, attendance, a cloud dashboard, camera-based vision, Wi-Fi sensing and offline recovery. Each level extends the previous one; earlier functions must continue working as new features are added.



Problem Statement

Why Do We Need a Smart Classroom?

- Traditional classrooms have limited real-time monitoring facilities.
- Manual attendance takes time and may cause errors.
- Fire and emergency situations need quick detection and response.
- Monitoring classroom occupancy and student activities manually is difficult.
- Network or power failures can interrupt normal classroom operations.

Overall System Objectives

Environmental Monitoring
Monitor temperature, humidity, heat index and fire conditions.

Safety & Security
Keep the classroom safe and handle emergencies.

Cloud Connectivity
Provide secure Wi-Fi, real-time data and cloud records.

Role-Based Access
Give Admin, Teacher, Student and Guardian limited access.

Smart Identification
Verify students using RFID card and face detection.

AI & Camera Vision
Detect occupancy, hand raise and monitor live view.

Presence Detection
Use Wi-Fi CSI for presence and room-zone localization.

Reliable Operation
Stay accessible during network outages and power cuts.

Remote Management
Support settings, firmware updates, weather notices, reports and AI assistant.

System Overview

How Does the System Work?

- ESP32-S3-CAM works as the main controller of the system.
- Sensors collect classroom environmental information.
- RFID and camera systems support student identification.
- Wi-Fi sends device information to the server.
- The web application displays real-time classroom information.
- Admin and Teachers can monitor and manage the system through the dashboard.

Hardware Components

Main Hardware Requirements

- ESP32-S3-CAM: Main controller with camera, Wi-Fi and Bluetooth.
- DHT11 Sensor: Measures classroom temperature and humidity.
- IR Flame Sensor: Detects fire conditions.
- RC522 RFID Reader: Reads student ID cards.
- RGB LED: Indicates the current system status.
- Active Buzzer: Provides alarm and warning sounds.
- Jumper Wires: Used for hardware connections and panic touch pad.

Environmental Monitoring

Temperature, Humidity and Fire Detection

- Continuously monitors classroom temperature and humidity.
- Calculates the heat index to understand classroom heat conditions.
- Detects fire using an IR flame sensor.
- Displays Safe, Warning and Danger conditions.
- Sends sensor information to the web dashboard.
- Provides alerts when abnormal conditions are detected.

Classroom Safety

Door Control and Emergency Response

- Maintains the simulated classroom door status as LOCKED or UNLOCKED.
- Automatically responds to fire detection.
- Keeps the fire condition active until an Admin or Teacher resets it.
- Includes a touch-based panic pad for emergency situations.
- Activates LED and buzzer alerts during emergencies.
- Displays FIRE and PANIC alerts on the dashboard.

User Management

Role-Based Access Control

The system provides different access permissions for four types of users.

- Admin: Manages users, system settings, classroom information and access permissions.
- Teacher: Monitors classroom conditions, attendance, safety alerts and student activities.
- Student: Accesses personal attendance and permitted classroom information.
- Guardian: Views linked children's arrival information and attendance histor

Student Identification

RFID and Face Detection

- Students use RFID cards for classroom check-in.
- The camera verifies student identity through face detection.
- Combines RFID card information with face verification.
- Records student attendance with entry time.
- Helps prevent unauthorized card usage.
- Provides attendance information through the web dashboard.

AI Camera Vision

Classroom Monitoring and Activity Detection

- Counts the number of people inside the classroom.
- Displays occupied and unoccupied seat zones.
- Provides live classroom viewing for Admin and Teachers.
- Detects unauthorized activity during Night Watch mode.
- Detects when the camera is covered or moved.
- Identifies students raising their hands using a trained AI model.

Wi-Fi CSI Detection

Presence Detection and Room Localization

- Uses Wi-Fi Channel State Information (CSI) to detect human presence.
- Identifies whether the classroom is empty or occupied.
- Estimates a person's location inside the classroom.
- Divides the classroom into a 3×3 grid.
- Works without depending on camera vision.
- Sends detection information to the server for monitoring.

Cloud Dashboard

Web-Based Monitoring and Management

- Displays real-time classroom temperature and humidity.
- Shows fire status, door status and system connectivity.
- Provides student attendance and recent card activity.
- Displays classroom occupancy and seat information.
- Stores sensor readings and system events in the database.
- Allows authorized users to monitor classroom information remotely.
- Provides device health information and system reports.

System Reliability

Temperature, Humidity and Fire Detection

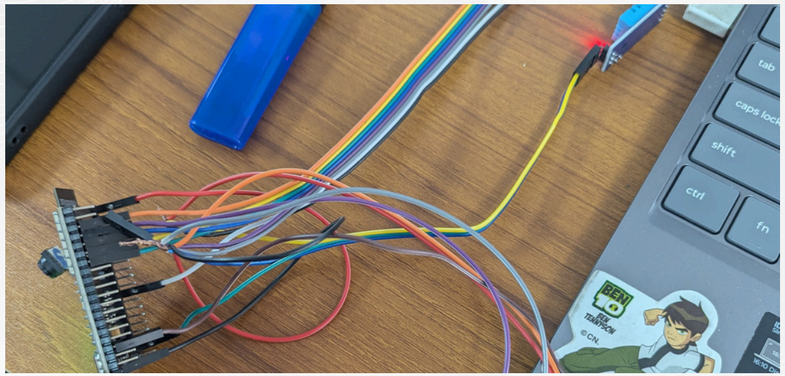
- Maintains important classroom functions during Wi-Fi disconnection.
- Provides a local access point when Wi-Fi is unavailable.
- Supports offline student entry using saved card information.
- Stores pending records locally during network failure.
- Synchronizes stored records after reconnection.
- Recovers safely after power interruptions.
- Prevents duplicate records during data synchronization.

Results

- Successfully monitors classroom temperature and humidity.
- Detects fire and generates emergency alerts.
- Identifies students using RFID and face verification.
- Records student attendance automatically.
- Displays classroom occupancy through AI camera vision.
- Detects student presence using Wi-Fi CSI.
- Shows real-time classroom information on the web dashboard.
- Stores classroom data for future monitoring and reports.
- Supports basic operations during network interruptions.

Conclusion

- Smart Classroom combines IoT, AI and web technologies.
- It improves classroom safety through real-time monitoring.
- It makes student identification and attendance more efficient.
- AI camera vision and Wi-Fi sensing provide smart classroom monitoring.
- Cloud connectivity allows remote access and management.
- Overall, the system creates a safer, smarter and more reliable classroom environment.

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Thank You